

Thomas Algorithm

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Tridiagonal System:

$$\begin{bmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

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$$\begin{bmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} b_1 & c_1 & & & \\ & \textcolor{red}{b'_2} & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ \textcolor{red}{d'_2} \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

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$$\begin{bmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{pmatrix}$$

Forward Elimination:

$$\begin{bmatrix} b_1 & c_1 & & & \\ & \color{red}{b'_2} & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} d_1 \\ \color{red}{d'_2} \\ d_3 \\ \vdots \\ d_n \end{pmatrix}$$
$$\begin{aligned} b'_2 &= b_2 - \frac{a_2}{b_1} \cdot c_1 \\ d'_2 &= d_2 - \frac{a_2}{b_1} \cdot d_1 \end{aligned}$$

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$$\begin{bmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} b_1 & c_1 & & & \\ & \color{red}{b'_2} & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ \color{red}{d'_2} \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$
$$\begin{aligned} b'_2 &= b_2 - \frac{a_2}{b_1} \cdot c_1 \\ d'_2 &= d_2 - \frac{a_2}{b_1} \cdot d_1 \end{aligned}$$
$$\begin{bmatrix} b_1 & c_1 & & & \\ & \color{red}{b'_2} & c_2 & & \\ & & \color{red}{b'_3} & c_3 & \\ & & & \ddots & \ddots \\ & & & & \color{red}{b'_n} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ \color{red}{d'_2} \\ \color{red}{d''_3} \\ \vdots \\ \color{red}{d_n^{(n-1)}} \end{Bmatrix}$$

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$$\begin{bmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} b_1 & c_1 & & & \\ & \color{red}{b'_2} & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ \color{red}{d'_2} \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$
$$\begin{aligned} b'_2 &= b_2 - \frac{a_2}{b_1} \cdot c_1 \\ d'_2 &= d_2 - \frac{a_2}{b_1} \cdot d_1 \end{aligned}$$
$$\begin{bmatrix} b_1 & c_1 & & & \\ & \color{red}{b'_2} & c_2 & & \\ & & \color{red}{b'_3} & c_3 & \\ & & & \ddots & \ddots \\ & & & & \color{red}{b'_n} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ \color{red}{d'_2} \\ \color{red}{d''_3} \\ \vdots \\ \color{red}{d_n^{(n-1)}} \end{Bmatrix}$$

Back Substitution:

$$x_n = \frac{d_n^{(n-1)}}{b'_n}$$

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Tridiagonal System:

$$\begin{bmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} b_1 & c_1 & & & \\ & \color{red}{b'_2} & c_2 & & \\ & a_3 & b_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & a_n & b_n \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ \color{red}{d'_2} \\ d_3 \\ \vdots \\ d_n \end{Bmatrix} \quad \begin{aligned} \color{red}{b'_2} &= b_2 - \frac{a_2}{b_1} \cdot c_1 \\ \color{red}{d'_2} &= d_2 - \frac{a_2}{b_1} \cdot d_1 \end{aligned} \quad \begin{bmatrix} b_1 & c_1 & & & \\ & \color{red}{b'_2} & c_2 & & \\ & & \color{red}{b'_3} & c_3 & \\ & & & \ddots & \ddots \\ & & & & \color{red}{b'_n} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} d_1 \\ \color{red}{d'_2} \\ \color{red}{d''_3} \\ \vdots \\ \color{red}{d_n^{(n-1)}} \end{Bmatrix}$$

Back Substitution: $x_n = \frac{d_n^{(n-1)}}{b'_n} \rightarrow x_i = \frac{d_i^{(i-1)} - c_i \cdot x_{i+1}}{b'_i}; \quad i = n-1, n-2, \dots, 1$

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Example 03-02: Find root of the following system of equations.

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ 10 \\ 10 \\ 210 \end{pmatrix}$$

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Example 03-02: Find root of the following system of equations.

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ 10 \\ 10 \\ 210 \end{pmatrix}$$

Forward Elimination:

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$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ 10 \\ 10 \\ 210 \end{pmatrix}$$

Forward Elimination:

$$\begin{bmatrix} 2 & -1 & & \\ 1.5 & -1 & -1 & \\ -1 & 2 & -1 & \\ & -1 & 2 & \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ 35 \\ 10 \\ 210 \end{pmatrix}$$

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Example 03-02: Find root of the following system of equations.

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ 10 \\ 10 \\ 210 \end{pmatrix}$$

Forward Elimination:

$$\begin{bmatrix} 2 & -1 & & \\ \textcolor{red}{1.5} & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \textcolor{red}{35} \\ 10 \\ 210 \end{pmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ \textcolor{red}{1.5} & -1 & & \\ & \textcolor{red}{1.333} & -1 & \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \textcolor{red}{35} \\ \textcolor{red}{33.33} \\ 210 \end{pmatrix}$$

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$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ 10 \\ 10 \\ 210 \end{pmatrix}$$

Forward Elimination:

$$\begin{bmatrix} 2 & -1 & & \\ \color{red}{1.5} & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ 10 \\ 210 \end{pmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ \color{red}{1.5} & -1 & & \\ & \color{red}{1.333} & -1 & \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ \color{red}{33.33} \\ 210 \end{pmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ & \color{red}{1.5} & -1 & \\ & & \color{red}{1.333} & -1 \\ & & & \color{red}{1.25} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ \color{red}{33.33} \\ \color{red}{235} \end{pmatrix}$$

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$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ 10 \\ 10 \\ 210 \end{pmatrix}$$

Forward Elimination:

$$\begin{bmatrix} 2 & -1 & & \\ \color{red}{1.5} & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ 10 \\ 210 \end{pmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ \color{red}{1.5} & -1 & & \\ & \color{red}{1.333} & -1 & \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ \color{red}{33.33} \\ 210 \end{pmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ & \color{red}{1.5} & -1 & \\ & & \color{red}{1.333} & -1 \\ & & & \color{red}{1.25} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ \color{red}{33.33} \\ \color{red}{235} \end{pmatrix}$$

Back Substitution:

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Example 03-02: Find root of the following system of equations.

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ 10 \\ 10 \\ 210 \end{pmatrix}$$

Forward Elimination:

$$\begin{bmatrix} 2 & -1 & & \\ & \color{red}{1.5} & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ 10 \\ 210 \end{pmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ & \color{red}{1.5} & -1 & \\ & & \color{red}{1.333} & -1 \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ \color{red}{33.33} \\ 210 \end{pmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ & \color{red}{1.5} & -1 & \\ & & \color{red}{1.333} & -1 \\ & & & \color{red}{1.25} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ \color{red}{33.33} \\ \color{red}{235} \end{pmatrix}$$

Back Substitution:

$$\color{red}{x_4} = \frac{235}{1.25} = \color{red}{188}$$

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Example 03-02: Find root of the following system of equations.

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ 10 \\ 10 \\ 210 \end{pmatrix}$$

Forward Elimination:

$$\begin{bmatrix} 2 & -1 & & \\ \color{red}{1.5} & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ 10 \\ 210 \end{pmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ \color{red}{1.5} & -1 & & \\ & \color{red}{1.333} & -1 & \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ \color{red}{33.33} \\ 210 \end{pmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ \color{red}{1.5} & -1 & & \\ & \color{red}{1.333} & -1 & \\ & & \color{red}{1.25} & \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ \color{red}{33.33} \\ \color{red}{235} \end{pmatrix}$$

Back Substitution:

$$\begin{aligned} x_4 &= \frac{235}{1.25} = \color{red}{188} \\ x_3 &= \frac{33.33 - (-x_4)}{1.333} = \frac{33.33 - (-188)}{1.333} = \color{red}{166} \end{aligned}$$

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Example 03-02: Find root of the following system of equations.

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ 10 \\ 10 \\ 210 \end{pmatrix}$$

Forward Elimination:

$$\begin{bmatrix} 2 & -1 & & \\ \color{red}{1.5} & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ 10 \\ 210 \end{pmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ \color{red}{1.5} & -1 & & \\ & \color{red}{1.333} & -1 & \\ & -1 & 2 & \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ \color{red}{33.33} \\ 210 \end{pmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ \color{red}{1.5} & -1 & & \\ & \color{red}{1.333} & -1 & \\ & & \color{red}{1.25} & \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ \color{red}{33.33} \\ \color{red}{235} \end{pmatrix}$$

Back Substitution:

$$\begin{aligned} x_4 &= \frac{235}{1.25} = \color{red}{188} \\ x_3 &= \frac{33.33 - (-x_4)}{1.333} = \frac{33.33 - (-188)}{1.333} = \color{red}{166} \\ x_2 &= \frac{35 - (-x_3)}{1.5} = \frac{35 - (-166)}{1.5} = \color{red}{134} \end{aligned}$$

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Example 03-02: Find root of the following system of equations.

$$\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ 10 \\ 10 \\ 210 \end{pmatrix}$$

Forward Elimination:

$$\begin{bmatrix} 2 & -1 & & \\ \color{red}{1.5} & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ 10 \\ 210 \end{pmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ \color{red}{1.5} & -1 & & \\ & \color{red}{1.333} & -1 & \\ & & -1 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ \color{red}{33.33} \\ 210 \end{pmatrix} \quad \begin{bmatrix} 2 & -1 & & \\ \color{red}{1.5} & -1 & & \\ & \color{red}{1.333} & -1 & \\ & & \color{red}{1.25} & \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 50 \\ \color{red}{35} \\ \color{red}{33.33} \\ \color{red}{235} \end{pmatrix}$$

Back Substitution:

$$\begin{aligned} x_4 &= \frac{235}{1.25} = \color{red}{188} \\ x_3 &= \frac{33.33 - (-x_4)}{1.333} = \frac{33.33 - (-188)}{1.333} = \color{red}{166} \\ x_2 &= \frac{35 - (-x_3)}{1.5} = \frac{35 - (-166)}{1.5} = \color{red}{134} \\ x_1 &= \frac{50 - (-x_2)}{2} = \frac{50 - (-134)}{2} = \color{red}{92} \end{aligned}$$